

Mayank Dixit

(412) 295-8008 | mavankd@andrew.cmu.edu | mayank-portfolio-gold.vercel.app | linkedin.com/in/mayank-dixit-max007

ML engineer and quantitative analyst in energy — deployed forecasting systems at a power-trading firm, production ML for utility-scale renewables, three peer-reviewed publications.

SKILLS

Languages & Tools: Python, SQL, MATLAB, JavaScript/TypeScript, Git, Docker

ML / DL: scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, Hugging Face; LSTM/GRU, TFT, Vision Transformers, RAG (FAISS + Llama)

Cloud & Data: AWS, Azure, GCP, PostgreSQL, SQL Server

Methods: time-series forecasting, uncertainty quantification (conformal, quantile), LLM agent pipelines (MCP), adversarial robustness

EDUCATION

Carnegie Mellon University — M.S., Artificial Intelligence Engineering — GPA 3.92/4.0

Dec 2026

Coursework: Machine Learning & AI for Engineers, Deep Learning, Systems & Toolchains for AI Engineers, Energy Conversion & Supply, Energy Transport & Storage

NSUT, Delhi, India — B.Tech., Electrical Engineering; Minor: ML & Data Science — CGPA 8.04/10

2019 – 2023

EXPERIENCE

Five Dimensions Energy — Quantitative Analyst Intern, Load Forecasting, Princeton, NJ

Jun – Aug 2026

- Improved the day-ahead load forecast feeding PJM auction bids for 16 utilities: regime-gated thermal build-up, humidity- and weekday-cooling interactions, and holiday weather calibration cut MAPE ~0.24 pts on 15 of 16 utilities across two independent held-out years; merged for the September production cycle.
- Built and deployed a real-time forecast for all 20 forecastable PJM zones — hourly, 14-day horizon, refreshed 4x/day with 90% uncertainty bands — outperforming the grid operator's own published forecast in every zone; per-horizon training approved and rolled out fleet-wide.
- Audited — unprompted — whether the production forward-forecast path built the same features as the validated backtests; it did not, and three defects were fixed before go-live with proof batteries (byte-identical no-op checks, training/backtest parity), then verified across every servable utility-class combination.
- Ran ~15 pre-registered experiments and a ~1,500-configuration deep-learning bake-off (LSTM/GRU/TFT) on dedicated compute — a definitive null against gradient-boosted trees; roughly two-thirds of tested ideas rejected on evidence, negatives documented as cleanly as wins.
- Built an LLM news-extraction pipeline for data-center capacity events (structured extraction, two-layer dedup, human-in-the-loop by design, 161 tests) and a facility-to-zone mapping validated 120/120 against independent regulatory records.

Carnegie Mellon University — Graduate Research Assistant — Prof. Christopher McComb

Sep 2025 – Present

- Architecting a Boeing-funded agentic-AI pipeline (open source: github.com/cmudrc) in which local open-weight LLMs — Gemma via Ollama, native function calling — autonomously run aircraft-design physics (geometry → CFD → propulsion → mission) across six MCP servers on a shared, version-tracked CPACS backbone; a multimodal verifier grounds each verdict in rendered CFD output.
- Built and deployed a production multi-agent AI coach (audio + text) on AWS delivering real-time team decision-making guidance in Slack, grounded in the lab's collective-intelligence research.

Gentari (PETRONAS) — Data Scientist → Senior Data Scientist, Gurgaon, India

Jul 2023 – Jul 2025

- Challenged third-party solar forecasts: engineered and productionized a bidirectional LSTM with novel exogenous features, surpassing vendor accuracy and yielding ~\$1M in annual savings.
- Architected and launched a private RAG chatbot (FAISS + local Llama 3.2), accelerating analyst knowledge retrieval and cutting API spend; led a Microsoft-partnered solar-forecasting framework on Azure (TFT, deep MNC) while mentoring two interns.
- Automated power-injection uploads to government portals for 40 solar/wind plants via Selenium (~100 man-hours saved weekly); deployed an LSTM inverter fault-detection model across company solar sites.

Netaji Subhas University of Technology — Research Assistant (B.Tech thesis), Delhi, India

Aug 2022 – Jun 2023

- Converted bearing vibration signals into time-frequency spectrograms (complex frequency B-spline) and trained NFNet and Vision Transformer classifiers (~98.7% validation accuracy on CWRU), producing two peer-reviewed publications.

Indian Institute of Technology Kharagpur — Research Intern, Kharagpur, India

May – Nov 2022

- Built a vision-based autonomous-car prototype (fine-tuned ResNet18, 99.6% sign detection; Raspberry Pi + Arduino control); red-teamed adversarial attacks (FGSM/BIM/PGD) and designed a structure-based self-supervised defense (avg. robustness 0.995) — published at CVIP 2022.

PUBLICATIONS

- Structure-Based Learning for Robust Defense Against Adversarial Attacks in Autonomous Driving Agents — CVIP 2022 (Springer)
- An NFNet-Based Approach for Enhanced Bearing Fault Classification Using Vibrational Signal Analysis — ICCNT 2023 (IEEE)
- TransForm-Motor: A Vision Transformer Transfer-Learning Framework for Bearing Fault Detection — ICAET (accepted)

ACHIEVEMENTS

- MIT Energy Hackathon 2025 — built Grid Scout for RWE: a U.S. data-center siting tool co-optimizing grid congestion headroom, clean-energy proximity, reliability, and emissions (grid-scout.vercel.app).
- 1st place of 110 teams, IIT Ropar Hackathon 2021 (Agri.AI, end-to-end ML platform for farmers); Indian Air Force Benevolent Association national scholarship for academic excellence.